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Comparing Soy and Milk Protein Regulation of Hepatic Omega-3 Fatty Acid Biosynthesis

Melissa Gonzalez-Soto¹ | Siobhan E. Woods¹ | Blair W. MacLeod¹ | James N. Smorenburg¹ | Geneviève J. DesOrmeaux¹ | Graham P. Holloway¹ | Ruxandra D. Rotarescu² | Miranda R. Green² | Adam H. Metherel² | David M. Mutch¹ 

¹Department of Human Health & Nutritional Sciences, University of Guelph, Guelph, Ontario, Canada | ²Department of Nutritional Sciences, University of Toronto, Toronto, Ontario, Canada

Correspondence: David M. Mutch (dmutch@uoguelph.ca)

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ABSTRACT

The omega-3 fatty acids eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) have numerous benefits, including strong anti-inflammatory and triglyceride-lowering properties. EPA and DHA are primarily obtained by consuming fatty fish; however, they are also endogenously synthesized primarily in the liver from α -linolenic acid (ALA) through a pathway mediated by the delta-6 desaturase (D6D) enzyme. Previous reports in rodents and humans suggest that dietary proteins such as soy and dairy may impact this pathway differently. The primary aim was to investigate the effects of diets containing either soy or milk protein on the expression, abundance, and enzymatic activity of the desaturases and elongases regulating hepatic omega-3 fatty acid biosynthesis. Male C57BL/6N mice ($n = 16$ per group) were fed a moderate-fat diet for 8 weeks containing either 1% or 3% energy from ALA. Protein content (15% energy) corresponded to either skim milk powder (SMP) or soy protein isolate (SPI). Hepatic fatty acid content was quantified by gas chromatography-flame ionization detection. Gene expression and protein expression were assessed by RT-qPCR and western blotting, respectively. D6D activity was measured in isolated hepatic microsomes. Fat oxidation was examined using a high-resolution respirometer. Hepatic omega-3 fatty acids (ALA, SDA, EPA, DPAn-3) were lower in SPI-fed mice compared to SMP-fed mice. *Fads1*, *Fads2*, *Elovl2*, and *Elovl5* expression was higher in SPI-fed mice compared to those fed SMP, while *Srebp-1c* expression was lower and *Cpt1a* expression higher in SPI-fed mice. Consistent with the changes seen at the gene expression levels, FADS2 protein abundance was higher in SPI-fed mice, whereas ELOVL5 protein expression was lower in the SPI groups. Little to no differences in microsomal D6D activity and mitochondrial respiration were detected. Our findings suggest that SPI-related reductions in hepatic omega-3 fatty acid content occur independent of changes in desaturase gene expression, protein expression, enzymatic activity, or mitochondrial respiration. Further studies investigating the influence of dietary proteins on ALA metabolism are therefore warranted.

Abbreviations: ALA, α -linolenic acid; D5D, delta-5 desaturase; D6D, delta-6 desaturase; DHA, docosahexaenoic acid; ELOVL2, elongation of very long chain fatty acids protein 2; ELOVL5, elongation of very long chain fatty acids protein 5; EPA, eicosapentaenoic acid; FADS1, fatty acid desaturase 1; FADS2, fatty acid desaturase 2; FAMES, fatty acid methyl esters.; GLA, γ -linolenic acid; LA, linoleic acid; LC-PUFA, long chain-polyunsaturated fatty acid; SDA, stearidonic acid.

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1 | Introduction

Omega-3 long-chain polyunsaturated fatty acids (n-3 LCPUFA), namely eicosapentaenoic acid (EPA, 20:5n-3) and docosahexaenoic acid (DHA, 22:6n-3), have well-recognized benefits for cardiometabolic health, including strong anti-inflammatory [1] and triglyceride-lowering properties [2]. EPA and DHA are primarily obtained from the consumption of fatty fish or over-the-counter fish oil supplements; however, EPA and DHA can also be endogenously synthesized from the essential fatty acid, α -linolenic acid (ALA, 18:3n-3). The pathway controlling EPA and DHA biosynthesis is regulated by several fatty acid desaturase and elongase enzymes that are influenced by both non-modifiable (e.g., age, biological sex, and genetics) [3] and modifiable factors (e.g., diet) [4]. Of specific relevance to the present study is past evidence that dietary proteins may differentially influence the activity of this pathway.

Soy is a commonly consumed plant-based protein with a comparable quality to that of typical animal proteins such as milk, eggs, and beef [5]. Prior evidence has shown that soy consumption is associated with various health benefits, including a lower incidence of coronary heart disease, obesity, and metabolic dysfunction-associated steatotic liver disease (MASLD) [6]. The inclusion of soy protein in the diet was reported to have significant hypolipidemic properties through the reduction of triglycerides, total cholesterol, and low-density lipoprotein cholesterol [7, 8]. The beneficial effects on circulating lipids are mediated in part through the ability of soy protein (and/or soy-derived isoflavones) to simultaneously reduce *de novo* lipogenesis via the downregulation of hepatic sterol regulatory element-binding protein-1 (SREBP-1) [9, 10] and upregulation of mitochondrial β -oxidation [9, 11]. While comparatively less is known about the relationship between soy protein consumption and regulation of n-3 LCPUFA synthesis, a link is plausible given that SREBP-1 is a transcriptional regulator of the desaturase genes that control the conversion of ALA into EPA and DHA [12].

Briefly, ALA and linoleic acid (n-6 LCPUFA) are converted into more highly unsaturated fatty acids through a pathway regulated by the delta-6 desaturase (encoded by *Fads2*), elongase 5 (encoded by *Elovl5*), delta-5 desaturase (encoded by *Fads1*), and elongase 2 (encoded by *Elovl2*) enzymes (see [4] for more details). Past research suggests that the consumption of soy protein may reduce the activity of this pathway. For example, studies by Ikeda et al. [13] and Koba et al. [14] both demonstrated that liver microsomal delta-6 desaturase (D6D) activity, as determined by the conversion of linoleic acid (LA) into downstream n-6 LCPUFA, was significantly lower in rats fed a diet containing soy protein compared to rats fed a diet containing casein (a milk protein). More recently, we conducted a cross-sectional analysis in ~1000 young Canadian adults and reported that the intake of soy foods and beverages was associated with reduced desaturase indices compared to total dairy intakes [15]. While these studies support a relationship between soy protein and reduced desaturase pathway activity compared to dairy proteins, there are several notable limitations, such as the use of casein rather than a complete milk protein in rodent studies, and the use of desaturase indices in human studies. As such, continued investigation

into soy protein regulation of the hepatic LCPUFA biosynthesis pathway is warranted.

The present study aimed to explore the mechanisms by which soy protein may affect ALA desaturation compared to a complete milk protein. Our primary objective was to investigate the effects of dietary proteins on gene expression, protein abundance, and functional activity of key players in the hepatic LCPUFA biosynthesis pathway. A secondary objective examined the effects of dietary protein on mitochondrial respiration. Based on available evidence, we hypothesized that soy protein would suppress D6D activity and reduce hepatic levels of EPA and DHA compared to dairy proteins and stimulate mitochondrial fatty acid supported respiration.

2 | Methods

2.1 | Animal Care and Feeding

10-week-old male C57BL/6N mice were obtained from Charles River Laboratories (Quebec, Canada). Mice were individually housed in conventionally ventilated shoebox cages at room temperature (22°C) in a humidity-controlled room with a 12:12-h light–dark cycle with *ad libitum* access to an AIN-93G diet (Teklad Laboratory Diets, IN, USA) and water for 2 weeks to allow for acclimatization. At 12 weeks of age, mice ($n = 16$ per diet) were randomly assigned to consume one of four isocaloric AIN-93G moderate-fat diets (Table S1; Research Diets, NJ, USA) modified to contain either a soy protein isolate (SPI; SUPRO661 from Solae LLC, MO, USA) or a complete milk protein (skim milk powder, SMP; CAT#902887 from MP Biomedicals, OH, USA) and either 1% (low-ALA) or 3% (high-ALA) energy from ALA (provided by flaxseed oil) for 8 weeks. The low-ALA diets are in alignment with the Acceptable Macronutrient Distribution Range (AMDR) for ALA [16]. The high-ALA diets were included to provide a higher level of precursor for the synthesis of EPA, as previous evidence showed a dose–response relationship between the amount of ALA provided and the amount of EPA produced [17]. Fatty acid content of diets was confirmed by gas chromatography–flame ionization detection (GC-FID). Diets provided 15% energy from protein (either SPI or SMP), 50% energy from carbohydrate, and 35% energy from fat. The macronutrient breakdown of experimental diets was chosen to align with the typical Western diet [18]. The SPI diet contained approximately 1.6 mg total aglycone isoflavones per g of protein isolate and was supplemented with calcium, lactose, methionine, and phosphorus to match the levels found in SMP to allow us to better investigate the specific impact of different dietary protein sources on n-3 LCPUFA biosynthesis. Body weight and food intake were measured weekly. All experimental protocols were approved by the University of Guelph Animal Care Committee in accordance with the requirements of the Canadian Council on Animal Care (Animal Utilization Protocol #4350).

2.2 | Tissue Collection

Prior to euthanasia, food was removed from cages in the morning and mice were fasted for 4 h prior to being anesthetized with

isoflurane. Blood was collected by cardiac puncture and animals were euthanized by decapitation. A subset of mice ($n=8$ per diet group) was euthanized by transcardiac perfusion with ice-cold saline solution for hepatic microsome isolation. Liver tissues were collected, weighed, sectioned for mitochondrial oxidation, or snap frozen in liquid nitrogen and stored at -80°C until further analysis.

2.3 | Lipid Extraction and Preparation of Fatty Acid Methyl Esters (FAMES)

Lipids were extracted from ~ 50 mg of liver tissue in a 2:1 chloroform: methanol using a modified Folch method [19]. After homogenization, lipid and aqueous phases were separated with the addition of 0.88% KCl, centrifuged at 500 g for 5 min, and the bottom lipid-containing chloroform phase was isolated. The resulting total lipid extracts were dried under a steady stream of nitrogen and transmethylated with 14% boron trifluoride in methanol for 1 h at 100°C to yield FAMES. The samples were cooled to room temperature, 1 mL of water and hexane were added, vortexed, and centrifuged at 500 g for 5 min. The top hexane layer containing FAMES was isolated, dried under nitrogen, re-suspended in 100 μL of heptane, and analyzed by GC-FID.

2.4 | GC-FID

FAMES generated from liver tissues were analyzed on a gas chromatograph equipped with a DB-FFAP 30 m $\times$ 0.25 mm i.d. $\times$ 0.25 mm film thickness capillary column (J&W Scientific from Agilent Technologies, Mississauga, ON). FAMES (1 μL) were introduced into the chromatograph by a Varian CP-8400 autosampler into the injector heated at 250°C with a split ratio of 30:1. The initial oven temperature was 50°C with a 1.0 min hold, followed by a $30^{\circ}\text{C}/\text{min}$ ramp to 130°C , a $10^{\circ}\text{C}/\text{min}$ ramp to 175°C , a $5.0^{\circ}\text{C}/\text{min}$ ramp to 230°C with a 9.5 min hold, and then a $50^{\circ}\text{C}/\text{min}$ ramp up to 240°C with an 11.1 min hold at the end. The FID temperature was 300°C , with air and helium gas flow rates of 300 and 25 mL/min, respectively, and a sampling frequency of 40 Hz. Peaks were identified by retention times through comparison to an external mixed standard sample (GLC-462, Nu Chek Prep). Fatty acid data is reported as relative % of total fatty acids.

2.5 | Reverse Transcription Quantitative Polymerase Chain Reaction (RT-qPCR)

Total RNA was extracted from liver tissue using TRIzol Reagent (CAT#15596026, Invitrogen, Massachusetts, USA) for tissue homogenization and the RNeasy Mini Kit (CAT#74106, Qiagen, Mississauga, ON, CA) for RNA extraction, as per the manufacturer's instructions. RNA concentration and purity were determined using a NanoDrop 2000 Spectrophotometer (Thermo Fisher Scientific, Wilmington, USA). All samples included in gene expression analyses had purity ratios (260/230 and 260/280) between 1.8 and 2.2. cDNA was synthesized from 1 μg of total RNA using the Applied Biosystems High-Capacity cDNA Reverse Transcription Kit (CAT#4368814, Thermo Fisher Scientific), as per the manufacturer's instructions. The primers used for RT-qPCR

were designed online using the Roche Universal Probe Library and Assay Design Center (Table S3). The total reaction volume per sample was 10 μL , which consisted of 5 μL SsoFast Evagreen Supermix (CAT#1725204, Bio-Rad Laboratories), 2.5 μL cDNA template, 0.2 μL of 10 μM forward and reverse primer mixture, and 2.3 μL nuclease-free water. RT-qPCR was run on a Bio-Rad CFX-96 Real-Time system with the following cycling conditions: one denaturing cycle at 95°C for 30s, followed by 40 cycles of 95°C for 4s and 55.9°C for 4s. All reactions were run in triplicate. 18s was used as a reference gene for normalization. Results were quantified using the $\Delta\Delta\text{Ct}$ method.

2.6 | Protein Extraction and Quantification

Liver tissue samples (30 mg) were homogenized in NP-40 cell lysis buffer (CAT#FNN0021, Invitrogen) containing a phosphatase and protease inhibitor cocktail (CAT#P178440, Thermo Scientific) and phenylmethylsulfonyl fluoride (PMSF). Tissue homogenates were centrifuged for 15 min at 1500 g at 4°C . Protein concentration was determined using a BCA assay. Protein samples were prepared using 2 $\times$ Laemmli buffer (CAT#1610737, BioRad) and denatured at 95°C for 5 min. 10 μg of protein was loaded and separated by SDS-PAGE prior to transfer to a nitrocellulose membrane (CAT#1620115, Bio-Rad Laboratories, California, USA) and blocked in 5% BSA or non-fat milk in Tris-buffered saline with Tween (TBST). The following primary antibodies were diluted in 5% non-fat milk unless otherwise indicated: FADS2 (CAT#28034-1-AP, ProteinTech, 1:250 dilution), FADS1 (CAT#126706, Abcam, 1:1000 dilution in 5% BSA), ELOVL2 (CAT#176327, Abcam, 1:1000 dilution), and ELOVL5 (CAT#TA338477, Origene, 1:500 dilution) for overnight incubation at 4°C . Membranes were washed in TBST and incubated in goat anti-rabbit horseradish-peroxidase-conjugated secondary antibody (1:2000 dilution; CAT#1706515, BioRad) in 5% BSA or non-fat milk in TBST for 2 h. Before imaging, membranes were incubated in Pierce ECL Western blotting substrate (CAT#32106, Thermo Scientific). Ponceau S solution (CAT#P7170, Millipore Sigma) was used as a loading control and for data normalization. Protein band intensities were quantified using AlphaView (Alpha Innotech Software, San Leandro, CA, USA) and normalized to the Ponceau S stain from the same membrane.

2.7 | Delta-6 Desaturase Microsomal Activity

Microsomes were isolated from liver tissue using a Microsomal Isolation Kit (CAT#ab206995, Abcam). Perfused liver (~ 300 mg) was homogenized with a dounce homogenizer. Liver homogenates were centrifuged at 10000 g for 15 min at 4°C ; the lipid layer was aspirated using a Pasteur pipette, and the supernatant was further centrifuged at 10000 g for 10 min at 4°C , followed by a third centrifugation at 20000 g for 20 min. The microsomal pellet was washed and resuspended in cold (4°C) storage buffer. Protein concentration was determined using a BCA assay.

Before initiating the reaction, ALA (CAT#90210, Cayman Chemical Company, MI, USA) was incubated with 2% fatty acid free BSA (CAT#A3803, Sigma) in PBS under constant stirring for 5 h at 37°C . For the enzymatic reaction, 1 mg/mL

of microsomal extract was incubated in media containing the following reagents: 4mmol ATP (CAT#A6419, Millipore Sigma), 0.1mmol coenzyme-A (CAT#C4780, Millipore Sigma), 1.28mmol NADPH (CAT#481973, Millipore Sigma), 0.5mmol nicotinamide (CAT#481907, Millipore Sigma), 5mmol $MgCl_2$ (CAT#7786-30-3, Millipore Sigma) and 2.42mmol N-acetylcysteine (CAT#A15409.14, Thermo Scientific) for 1 h in a shaking water bath at 37°C. BSA-bound ALA (10 μ M) was added as a substrate for D6D. After 1 h, the reaction was stopped by the addition of 2:1 methanol: chloroform with 0.005% of butylated hydroxytoluene (CAT#101162, MP Biomedicals, OH, USA). Total fatty acids were quantified by GC-FID, as described above.

2.8 | High Resolution Respirometry

Mitochondrial respiration was performed in a high-resolution respirometer (Oroboros Oxygraph 2-K; Oroboros Instruments, Innsbruck, Austria), as previously described [20, 21]. Briefly, a fresh portion of the liver was placed in ice-cold BIOPS preservation buffer (50mM MES, 7.23mM K_2EGTA , 2.77mM CaK_2EGTA , 20mM imidazole, 0.5mM dithiothreitol, 20mM taurine, 5.77mM ATP, 15mM PCr, and 6.56mM $MgCl_2 \cdot H_2O$;

pH 7.1), sectioned in small strips (~2–4mg wet weight) and transferred to ice-cold MiR05 (0.5mM EGTA, 3mM $MgCl_2 \cdot H_2O$, 60mM potassium lactobionate, 10mM KH_2PO_4 , 20mM HEPES, 110mM sucrose, 20mM taurine, and 1g L^{-1} fatty acid-free BSA; pH 7.1). Each respirometry chamber containing 2mL MiR05 was set to 37°C, and stirring was maintained at 750 rpm. Mitochondrial respiration was determined following the sequential addition of 5mM pyruvate and 2mM malate (PM), 5mM ADP (+D), 10mM glutamate (+G), and 10mM succinate (+S). Palmitoyl-CoA (PCoA) supported respiration (5 μ M and 40 μ M PCoA) was assessed in the presence of 2mM malate and 5mM ADP (MD) and 30 μ M L-Carnitine (+LC). In a separate lipid-supported respiration, the inhibitory effects of malonyl-CoA (MCoA, 1, 3, 5 and 8 μ M) on carnitine palmitoyl transferase 1 (CPT1) were determined in the presence of 2mM malate, 5mM ADP, 30 μ M L-Carnitine, and 5 μ M PCoA.

2.9 | Statistical Analyses

Data were analyzed using GraphPad Prism, version 10.0.2 (GraphPad Software Incorporated, La Jolla, California, USA). Normality was assessed with a Shapiro–Wilk test. Differences

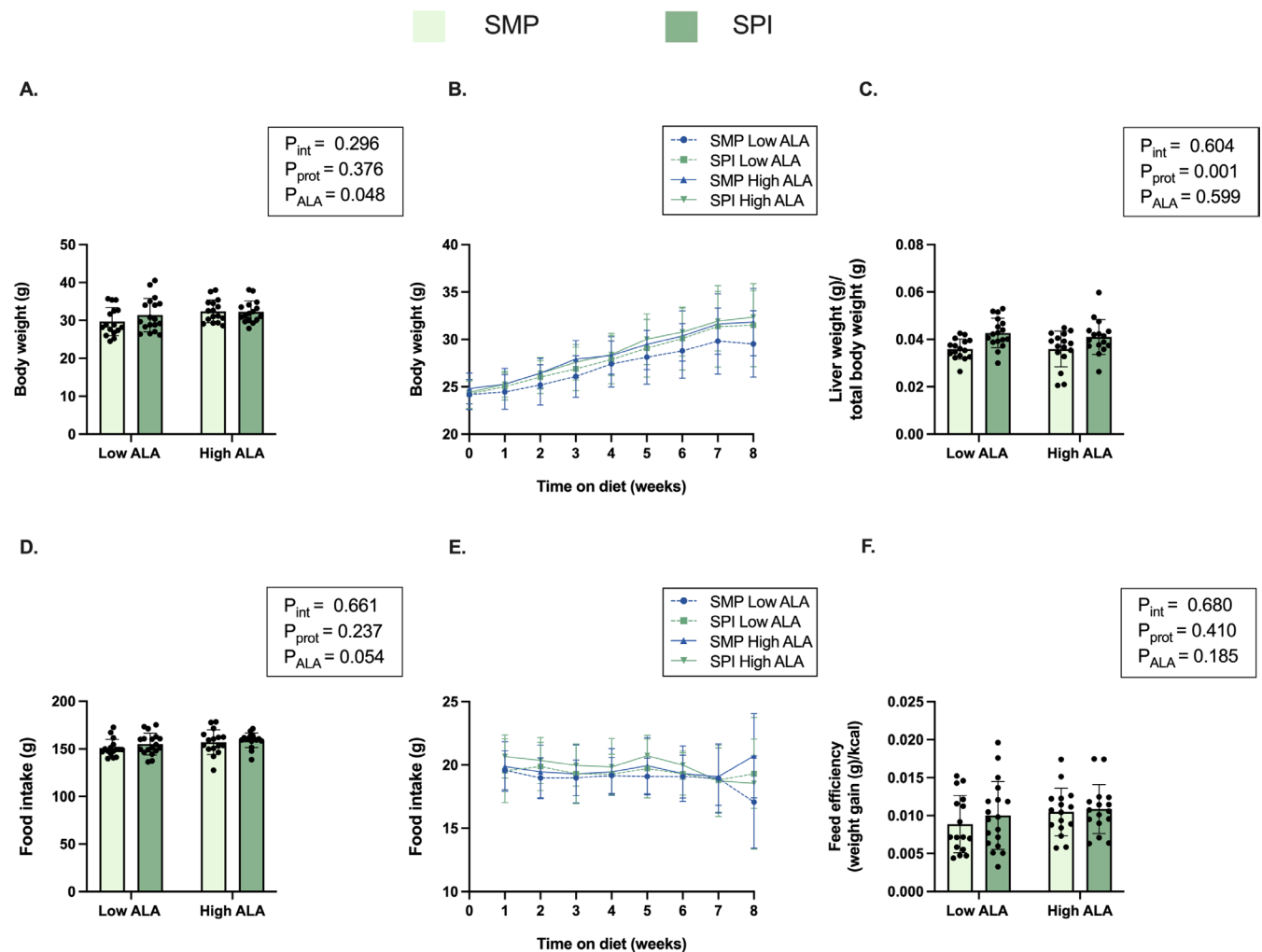


FIGURE 1 | Body weight and food intake measurements in mice fed experimental diets. (A) Total body weight, (B) weekly body weight, (C) relative liver weight, (D) total food intake, (E) weekly food intake, and (F) feed efficiency in mice fed experimental diets for 8 weeks are shown. Data are reported as mean $\pm$ SD. $n = 16$ per group.

between experimental groups were analyzed using a two-way analysis of variance (ANOVA) with dietary ALA and protein as the main factors, followed by an uncorrected Fisher's LSD test. A three-way ANOVA was used to assess differences in body weight and food intake over the 8-week study, with dietary ALA, protein, and time as the main factors. All data are presented as mean $\pm$ SD. A $p < 0.05$ was considered statistically significant.

3 | Results

3.1 | Body Weight, Relative Liver Weight, and Food Intake Are Minimally Affected by Dietary Protein and ALA Content

Mice fed high-ALA diets had slightly greater body weights compared to mice fed low-ALA diets ($P_{\text{ALA}} = 0.048$, Figure 1A), which aligned with a trend for higher weekly body weight during the study ($P_{\text{ALA}} = 0.082$, Figure 1B). In contrast, relative liver weight was higher in SPI-fed mice regardless of ALA content ($P_{\text{prot}} < 0.001$, Figure 1C). Total food intake tended to be higher in ALA-fed mice ($P_{\text{ALA}} = 0.054$, Figure 1D,E). No differences were observed in feed efficiency (Figure 1F).

3.2 | Hepatic n-3 PUFA Profiles Differ With High-ALA Diets, but Not Low-ALA Diets

Hepatic fatty acid profiles were affected by both dietary protein and ALA content (Table S2). Total levels of n-3 PUFA reflected dietary ALA content (main effects for protein and ALA both $p < 0.001$, Figure 2A–D), with mice fed high-ALA diets

having a greater proportion of n-3 PUFA in the liver than mice fed low-ALA diets. Total n-3 PUFA was also lower in mice fed SPI diets compared to those fed SMP diets ($P_{\text{prot}} = 0.003$; Figure 2). While total monounsaturated fatty acid (MUFA) levels appeared to be higher in low ALA-SPI fed mice, this difference was not significant ($P_{\text{prot}} = 0.116$). Additionally, no differences in total saturated fatty acid (SFA) and total n-6 PUFA were observed in the liver.

We next examined hepatic levels of individual n-3 fatty acids in mice (Figure 3A–E). Relative levels of hepatic ALA (Figure 3A), stearidonic acid (SDA, 18:4n-3; Figure 3B), EPA (Figure 3C) and docosapentaenoic acid (DPAn-3, 22:5n-3; Figure 3D) were elevated in mice fed the high-ALA diets compared to low-ALA diets, with no difference observed in DHA (Figure 3E). In the high-ALA SPI-fed mice, hepatic ALA levels were lower compared to high-ALA SMP mice (Figure 3A); however, this mirrored the ALA content of the diets (see small blue insert bar graphs in Figure 3), explaining the differences between groups. In contrast, SDA, EPA, and DPAn-3 levels were lower in mice fed the high-ALA SPI diet compared to those fed the SMP diet (Figure 3B–D). These changes did not reflect their levels in the various diets, suggesting a possible differential regulation of the LCPUFA biosynthesis pathway between diet groups.

3.3 | Expression of Hepatic n-3 LCPUFA Biosynthesis Genes and Corresponding Proteins Are Influenced by SPI Consumption

To further investigate the differences in hepatic n-3 PUFA levels in mice fed the various diets, we first measured the expression of hepatic desaturase and elongase genes. Regardless of

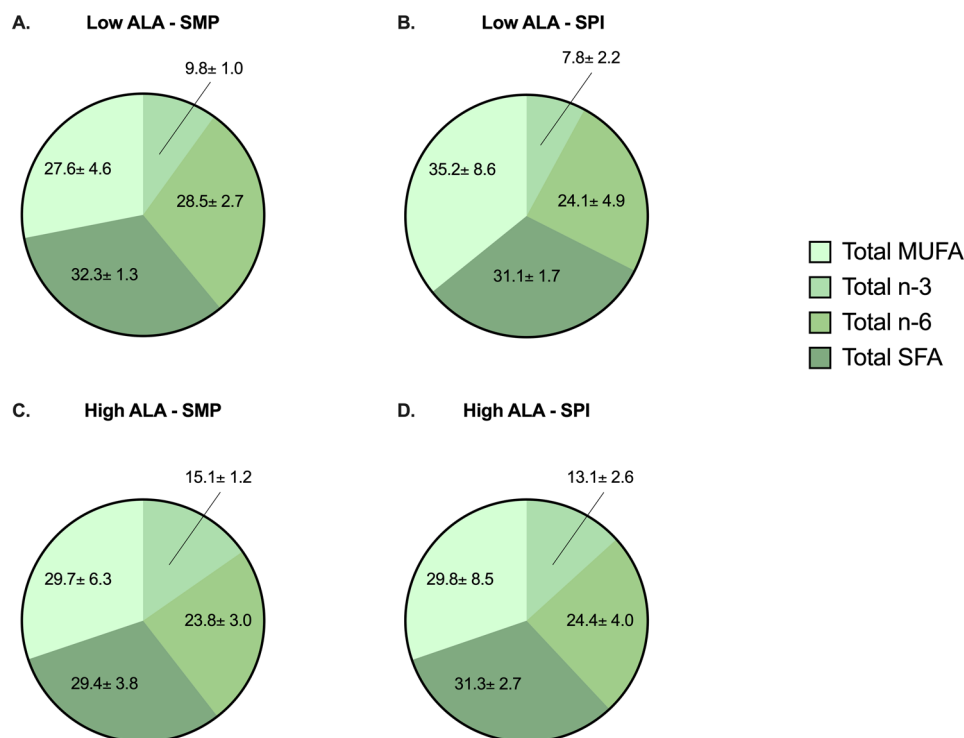


FIGURE 2 | Total fatty acid levels in liver. Pie charts depicting the percentage breakdown of fatty acid classes (SFA, MUFA, n-3, and n-6) in mice fed the different dietary groups are shown (A–D). Data are reported as mean $\pm$ SD (see Table S2 for values). $n = 9$ per group.

ALA content in the diet, mice consuming SPI showed higher *Fads2* (Figure 4A), *Fads1* (Figure 4B), *Elovl2* (Figure 4C), and *Elovl5* (Figure 4D) expression compared to SMP-fed mice. These mRNA results suggest that the lower levels of SDA and EPA seen in high-ALA SPI-fed mice compared to high-ALA SMP-fed mice are not attributed to reductions in the expression of key desaturases and elongases. Mice consuming SPI had lower *Srebp-1c* expression compared to those consuming SMP, which was particularly evident in the low ALA groups (Figure 4E). This suggests that *Srebp-1c* is not solely responsible for the regulation of desaturase gene expression. Finally, a main effect of protein was observed with *Cpt1a* expression (Figure 4F), where SPI-fed mice had higher expression than SMP-fed mice, regardless of ALA content.

We next explored if changes in desaturase and elongase mRNA were reflected at the protein level. A significant main effect of protein was observed for FADS2 and ELOVL5, a significant main effect of diet ALA content was found for ELOVL2, and no differences were found in FADS1 content (Figure 5A–D). Specifically, FADS2 protein content was increased with SPI, similar to *Fads2* mRNA (Figure 5A), while ELOVL5 protein content was reduced with SPI (Figure 5D). In contrast, no effect of dietary protein was observed on ELOVL2 protein

content; however, ELOVL2 content was marginally higher in high-ALA diet groups compared to low-ALA diet groups (Figure 5C).

3.4 | Dietary Protein Does Not Impact D6D Activity in Hepatic Microsomes

We next investigated D6D activity in isolated hepatic microsomes. Enzymatic activity ($\mu\text{g}/\text{mg}/\text{min}$) was determined by measuring the amount of SDA present in microsomes following incubation with ALA. Baseline levels of ALA and SDA in microsomes for the four groups of mice reflected that seen in whole livers (data not shown). No differences were observed in the production rate of SDA between the four groups (Figure 6).

3.5 | Mitochondrial Respiration Is Not Influenced by Dietary Protein or ALA Levels

Finally, we measured lipid-supported respiration as an exploratory secondary outcome to investigate if dietary protein impacted mitochondrial function, thus providing further insight into the lower levels of ALA, SDA, and EPA seen in mice fed the

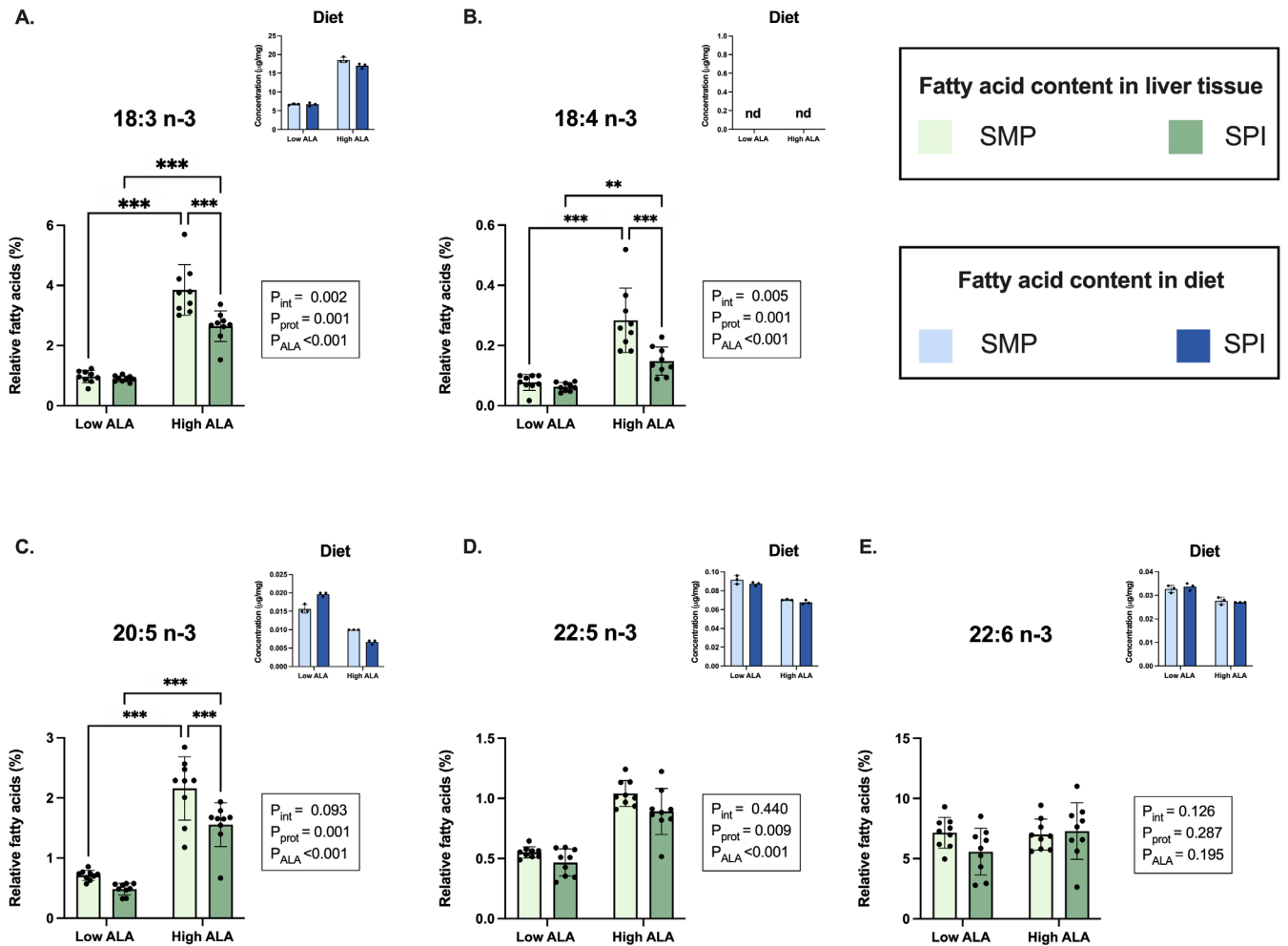


FIGURE 3 | Relative n-3 PUFA composition in liver. Hepatic levels of the main n-3 PUFA are shown (A–E). Small blue insert graphs depict the content of the same fatty acid in the four experimental diets. Post hoc test significance: *represents a $p \leq 0.05$, **represents a $p \leq 0.01$, and ***represents a $p \leq 0.001$. $n = 9$ per group.

high-ALA SPI diet compared to mice fed the high-ALA SMP diet (i.e., altered fat oxidation). Our analysis revealed that neither dietary protein nor ALA altered mitochondrial maximal oxidation capacity during complex-I supported oxidative phosphorylation (PMD+G) nor complex I and II-linked oxidative phosphorylation (PMDG + S) (Figure 7A). Similarly, no differences in PCoA-supported respiration were observed between groups (Figure 7B). In a separate protocol, we titrated MCoA (a biological CPT1 inhibitor) at various concentrations and observed a progressively greater effect of malonyl-CoA inhibition of PCoA oxidation at higher concentrations in relation to dietary protein, regardless of ALA content (Figure 7C). Overall, these results suggest that SPI may reduce the ability for mitochondria to oxidize fatty acids. However, given that reduced oxidation would be expected to result in greater accumulation of various fatty acids, these results cannot account for the decreased levels of omega-3 fatty acids observed following the consumption of SPI.

4 | Discussion

Epidemiological evidence supports the benefits of soy protein consumption for improving blood lipid profiles [7, 22, 23], with increasing evidence suggesting benefits in individuals with

MASLD as well as animal models [6]. Past studies have shown that soy protein influences *de novo* lipogenesis (DNL) through the downregulation of *Srebp-1c* expression and its target genes (*Fasn*, *Acc* and *Scd*) compared to casein [10, 24–26], while fatty acid oxidation is increased [9, 27]. However, the impact of dietary protein on n-3 LCPUFA metabolism is comparatively less explored. In the present study, we report that the consumption of soy-protein diets led to a reduction in hepatic n-3 LCPUFA content compared to a diet containing a complete milk protein. While the lower liver ALA may relate to differences in the fatty acid content of experimental diets, the reduction in more highly unsaturated n-3 LCPUFA suggested a potential influence on the biosynthesis pathway. However, our findings do not support that the differences in liver fatty acid profiles between SPI and SMP diets are related to changes in hepatic desaturase gene expression, protein abundance, or function.

Hepatic n-3 LCPUFA levels reflected the quantity of ALA in the different diets, consistent with previous studies showing that increased dietary ALA resulted in increased EPA and DPAn-3 levels in rodents [28, 29] and humans [30]. Interestingly, hepatic DHA levels were not different across the four groups of mice; however, this aligns with other studies showing that higher ALA intake does not change the size of the DHA pool,

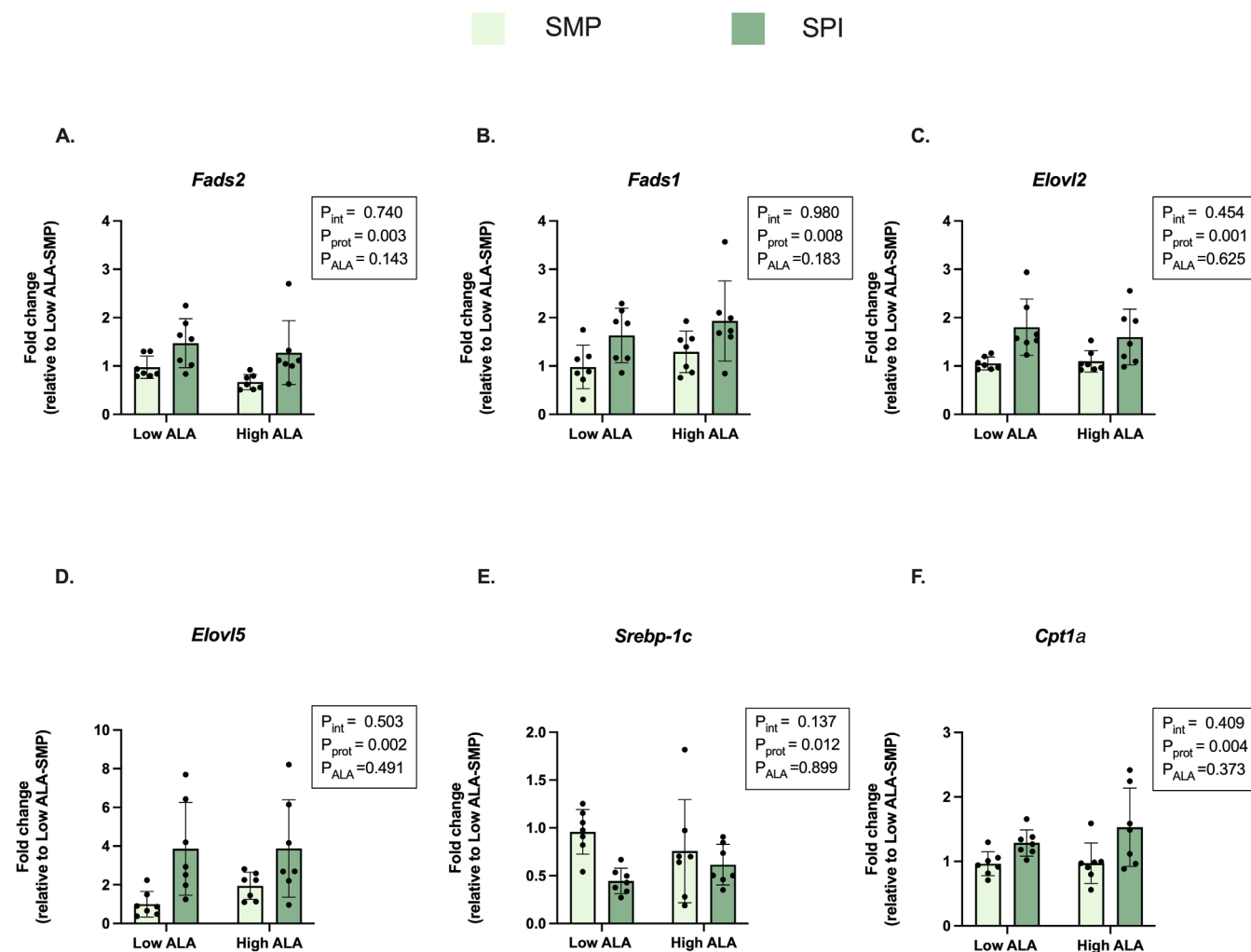


FIGURE 4 | Hepatic mRNA levels of genes involved in the biosynthesis pathway. (A) *Fads2*, (B) *Fads1*, (C) *Elovl2*, (D) *Elovl5*, (E) *Srebp-1c*, and (F) *Cpt1a*. Post hoc test significance: *represents a $p \leq 0.05$, **represents a $p \leq 0.01$, and ***represents a $p \leq 0.001$. $n = 7$ per group.

particularly in diets lacking pre-formed DHA [28, 31]. Indeed, Metherel et al. [32] reported a higher DHA synthesis-secretion coefficient from plasma unesterified EPA compared to ALA, which was attributed to EPA requiring fewer enzymatic steps for conversion into DHA. These same authors also reported that higher plasma ALA levels compensated for lower biosynthesis, leading to no significant differences in overall DHA levels. These findings are supportive of our results, as we also observed no changes in DHA levels when ALA was the primary source of n-3 PUFA in the diet.

To explore the potential mechanisms underlying reductions in hepatic SDA and EPA content, we examined gene and protein expression of enzymes involved in the desaturase pathway. Contrary to what we expected, mice fed SPI diets exhibited increased expression of desaturase and elongase genes (*Fads1*, *Fads2*, *Elovl2* and *Elovl5*), potentially as a compensatory response to lower ALA availability. Furthermore, these increases in gene expression were observed despite the reduction in *Srebp-1c* expression expected in mice fed soy protein (Figure 4E). However, changes in mRNA were not

fully reflected at the protein level. Indeed, only FADS2 (D6D) protein content was increased with soy protein. In contrast, ELOVL5 was reduced with soy protein while ELOVL2 content was higher in high-ALA diets regardless of protein content. Finally, a functional desaturase assay examining conversion of ALA into SDA (i.e., the step mediated by *Fads2*/D6D) in isolated liver microsomes showed no differences across the dietary groups. Taken together, this suggests that the small but significant changes in desaturase and elongase gene expression and protein content, while statistically significant, were not sufficient to translate to meaningful changes at the functional level, most probably due to inter-animal variability. Further functional assays with larger sample sizes or targeting D5D and elongase activities may reveal different points of regulation by soy and dairy proteins in the desaturase pathway.

Our results seemingly contrast with prior studies that reported lower D6D activity in rats fed SPI compared to those fed casein [13, 14]; however, there are several noticeable differences to highlight. First, these past studies assessed D6D activity by measuring the conversion of LA (18:2n-6) into γ -linolenic acid (GLA,

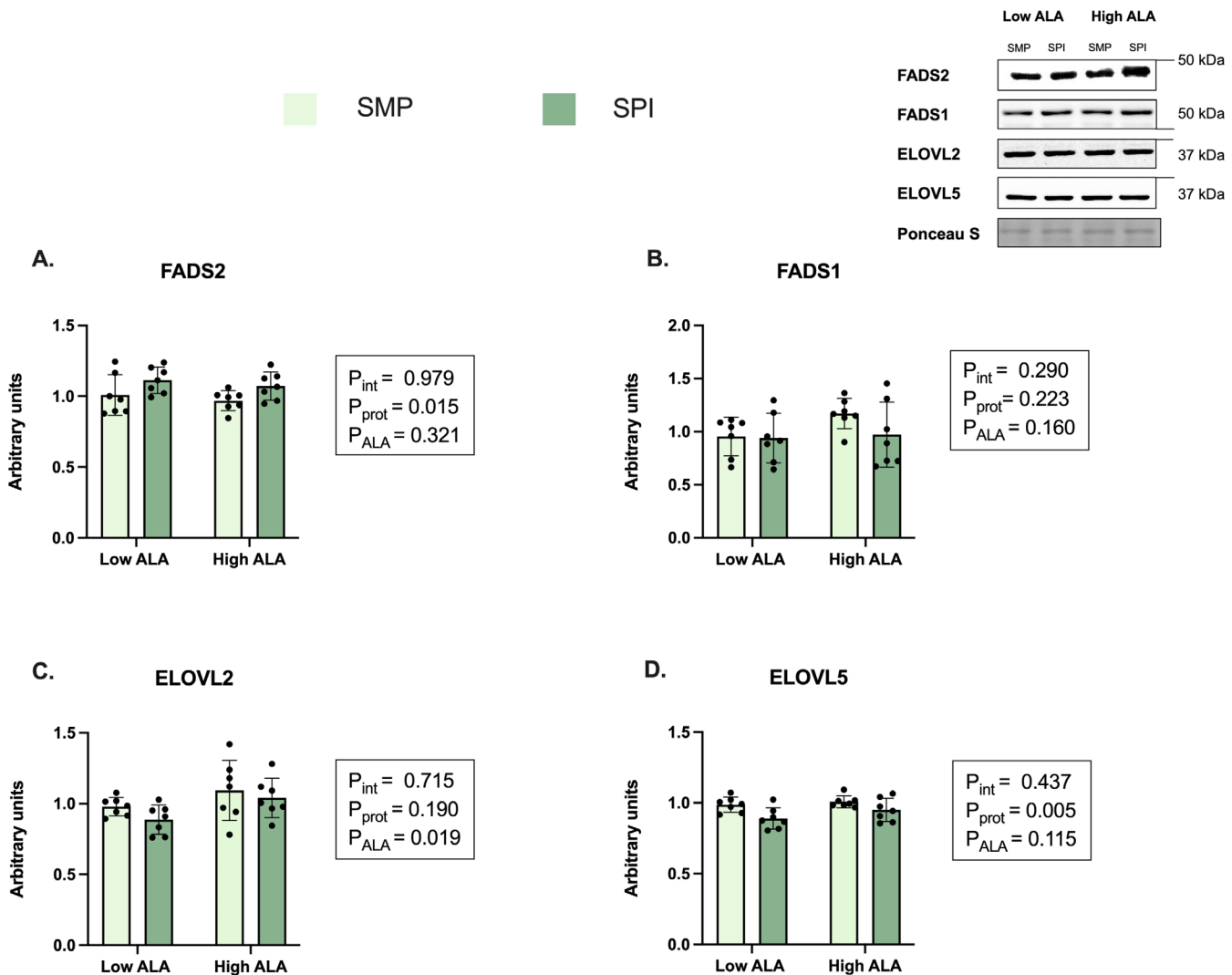


FIGURE 5 | Hepatic protein expression of desaturases and elongases. (A) FADS2 (D6D), (B) FADS1 (D5D), (C) ELOVL2, and (D) ELOVL5. Experimental groups are indicated at the top. The molecular weight is shown on the righthand side. Ponceau S was used as a loading control and for data normalization. $n = 7$ per group.

18:3n-6). Given the known competition that exists between ALA and LA for flux through this pathway [33], it is possible that discrepant results between studies may be attributed to differences in the fatty acid composition of experimental diets. Second, the effects of soy protein on desaturase pathway activity may be acute rather than chronic. For example, Ikeda et al. showed a reduction in D6D activity after 7 days of SPI feeding [13] which seemed to normalize at 14 days, suggesting that the effects of SPI consumption on D6D activity may be transient. In contrast, our results align with the outcomes of a small parallel-arm human trial suggesting that the short-term consumption of a soy protein beverage powder had no effect on plasma phospholipid fatty acid profiles or desaturase indices compared to casein [34]. Given the known effects of numerous nutrients on the regulation of the LCPUFA biosynthesis pathway [4], it is plausible that differences in background diets may provide some explanation to reconcile these differing results across the various studies. For

example, soy protein is limiting in methionine. To avoid potential issues related to this limiting amino acid, we supplemented all diets with an equivalent amount of L-methionine. However, Shimada et al. [35] reported that supplementing a 10% casein diet with 0.43% L-methionine significantly increased microsomal D6D activity compared to a control casein diet. To improve the translatability of our work, we chose to use a skim milk powder rather than a milk protein isolate or individual milk proteins (i.e., casein or whey); however, this meant that we had to adjust the SPI-containing diets for lactose and calcium content in order to focus specifically on the effects of protein. We cannot exclude that the addition of these other nutrients could have influenced our results and may contribute to the discrepancies observed with other studies. This is notable given that previous investigations have tended to compare soy protein with casein and/or whey on the LCPUFA synthesis pathway rather than skim milk powder. Additional studies examining the influence of individual nutrients on desaturase pathway activity are therefore warranted.

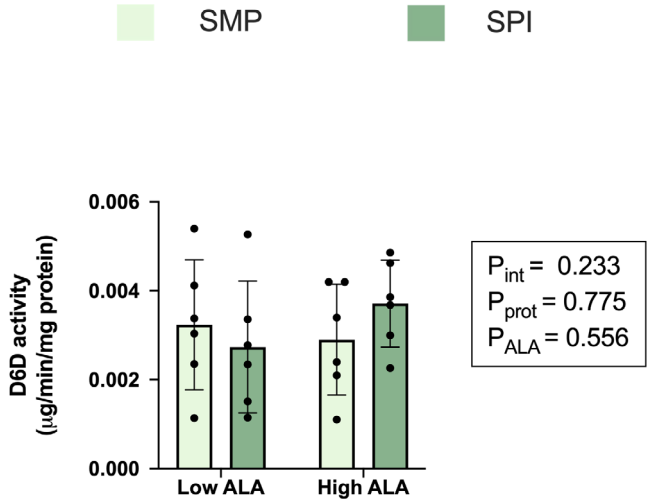


FIGURE 6 | Delta-6 desaturase activity in isolated hepatic microsomes. Microsomes were obtained from liver tissue and incubated with ALA. D6D activity is reported as $\mu\text{g}/\text{min}/\text{mg}$ of protein. $n=6$ per group.

In addition to ALA conversion into n-3 LCPUFA, ALA can also be partitioned toward β -oxidation [36]. We found that SPI-diets, irrespective of ALA content, increased *Cpt1a* expression; however, no major changes were seen in mitochondrial oxidative capacity. Although direct comparisons between soy and a complete milk protein are limited, several studies suggest that soy protein may enhance mitochondrial oxidation. For example, Xiao et al. [37] conducted a proteomic analysis on the livers of rats fed SPI or casein and reported an upregulation of several proteins involved in fatty acid oxidation in the SPI group. Similarly, Akahoshi et al. [38] observed increased CPT activity in rats fed SPI compared to casein. Finally, Takahashi et al. [39] reported a higher uncoupling protein 1 (*Ucp1*) expression in brown adipose tissue of SPI-fed rats, further supporting a potential role of SPI in fatty acid utilization. In our study, we did not observe any differences in mitochondrial oxidative capacity between dietary groups. However, a notable limitation with our analysis of mitochondrial respiration was the use of palmitoyl-CoA instead of ALA-CoA. It is therefore possible that performing these same

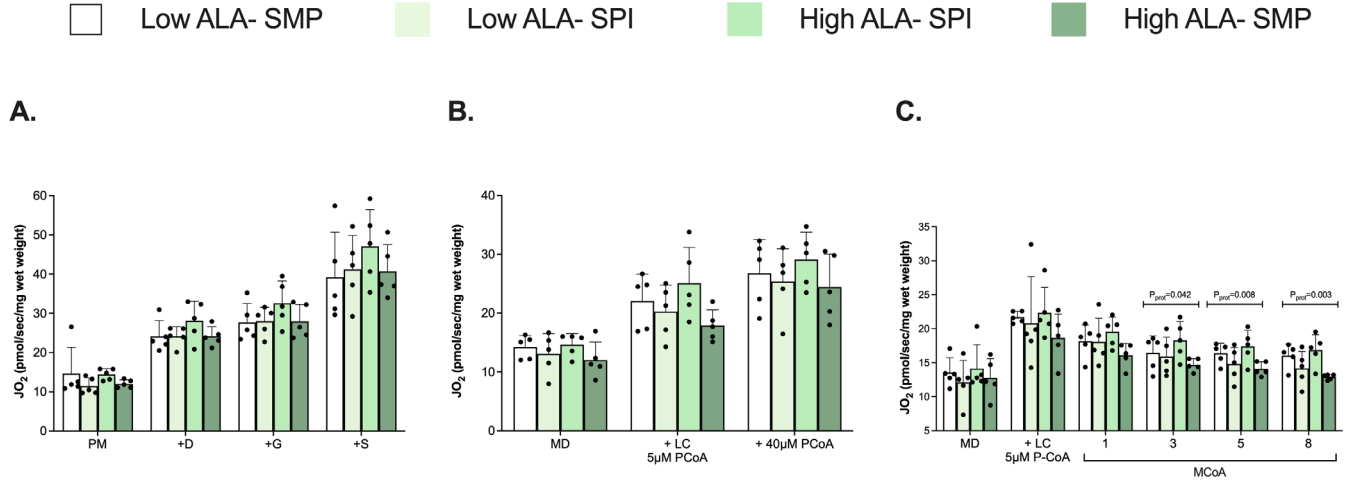


FIGURE 7 | Hepatic mitochondrial respiration. (A) Graphs for complex I and II supported respiration, (B) PCoA supported respiration, and (C) MCoA capacity to inhibit CPTI. PM, pyruvate + malate; +D, PM + ADP; +G, PMD + glutamate; +S, PMDG + succinate; MD, malate + ADP; +LC + PCoA, L-carnitine + palmitoyl-CoA; MCoA, malonyl-CoA. Statistical significance (main effect) is indicated with a horizontal line accompanied by the associated p -value. $n=5-6$ per group.

analyses with ALA-CoA may have a different outcome. This is relevant given that existing evidence shows a higher preference of CPT1 toward ALA when compared to other fatty acids [40]. Furthermore, we observed a significant effect of dietary protein on malonyl-CoA inhibition of palmitoyl-CoA oxidation. This suggests that soy protein may promote a reduced hepatic fatty acid oxidation capacity compared to a complete milk protein; however, this is difficult to interpret without directly measuring malonyl-CoA levels. A previous study by Gudbrandsen et al. [9] reported increased hepatic palmitoyl-CoA oxidation in obese Zucker rats fed an isoflavone-rich soy concentrate, with no change in malonyl-CoA levels or malonyl-CoA sensitivity of CPT1. Future studies examining mitochondrial respiration using ALA-CoA as a substrate are needed to determine if SPI may increase ALA oxidation specifically.

There are several considerations and future directions related to the present study to highlight. First, our study did not directly address ALA bioavailability and absorption in the gut. We previously reported that fecal energy content was higher in both male and female mice consuming the low-ALA SPI diet compared to the low-ALA SMP diet [41]. While fecal samples were not collected from mice fed high-ALA diets, this previous work suggests that energy extraction from the SPI diet is lower than from the SMP diet. Future studies in which the fatty acid profiles of fecal samples are assessed would provide specific insight into ALA absorption. Second, the use of a stable isotope or gas chromatography isotope ratio mass spectrometry (GC-IRMS) would complement the use of hepatic microsomes and provide greater clarity regarding ALA conversion into downstream n-3 LCPUFA. Third, we acknowledge that supplementing both the SPI and SMP with methionine may mask some of the effects previously reported regarding the impact of soy protein on desaturase activity. While the present study focused more on dietary ALA content in diets with equivalent amounts of methionine, future studies could compare diets with and without supplementation of this essential amino acid. Fourth, previous reports suggest that diets containing soy protein have lower circulating insulin levels than a diet containing casein [10, 12]. Future studies measuring circulating hormones could provide additional insights to better explain the discrepancy between *Srebp-1c* expression and that of desaturases and elongases. Finally, our study was conducted in male mice only. Although previous research has shown that hepatic conversion of ALA to DHA does not differ between sexes [42], it remains unclear whether dietary protein modulates this pathway differently in males and females.

5 | Conclusion

The present study shows that hepatic n-3 LCPUFA levels are lower in mice fed SPI compared to SMP. These reductions appear to occur independent of changes in desaturase gene expression, protein abundance, or activity, suggesting further investigation is necessary. Although we observed differences in malonyl-CoA inhibition of palmitoyl-CoA oxidation in our exploratory secondary analysis, the potential for increased ALA oxidation and reduced absorption efficiency remains to be explored. These findings highlight the need for additional studies to investigate the metabolic fate of ALA in the context of an SPI-based diet,

which will provide important insights into the broader implications of soy protein consumption on n-3 LCPUFA metabolism and overall health.

Author Contributions

M.G.-S. conducted experiments, analyzed the data, and drafted the original manuscript. S.E.W., B.W.M., and J.N.S. contributed to data collection, interpretation, and manuscript editing. G.J.D. conducted the mitochondrial assays, analyzed and interpreted the data, and contributed to manuscript editing. R.D.R. and M.R.G. conducted gas chromatography analysis and contributed to manuscript editing. G.P.H. and A.H.M. contributed to data analysis, interpretation, and manuscript editing. D.M.M. conceptualized the study design, contributed to the data analyses and interpretation, drafted the original manuscript, and acquired funding. All authors approved the final version of the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in the Results and [Supporting Information](#) of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.